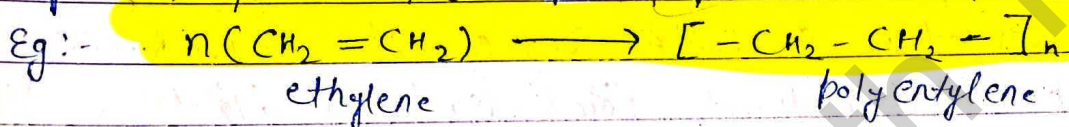


## CHAPTER - 15

# POLYMERS

**Monomers:-** The word polymer has a Greek origin, which means many units (parts). Polymer is defined as a chemical substance of a high molecular mass formed by the combination of a large number of simple molecules, called monomers.



**Polymerisation:-** The process by which the monomers get combined and transformed into polymers is known as polymerisation.  
 $n[\text{monomer}] \longrightarrow \text{Polymer}$

## CLASSIFICATION OF POLYMERS:-

On the basis of origin (source) of polymers:-

- 1) Natural polymers, e.g. proteins, cellulose, starch etc.
- 2) Semi-synthetic polymers e.g. Rayon
- 3) Synthetic polymers, e.g. polythene, buna-S, PVC, nylon, dacron etc.

On the basis of structure of polymers:-

- 1) Linear polymers e.g. polyvinyl chloride (PVC)
- 2) Branched chain polymers e.g. low density polythene
- 3) Cross-linked OR Network polymers e.g. bakelite, melamine

On the basis of mode of polymerisation:-

- 1) Addition polymers:- These are formed by the repeated addition



of monomer molecules possessing double or triple bonds. e.g. eg:- polythene is formed from ethene.

They are again classified as:-

A) Homopolymers: - These polymers are formed by the polymerisation of single monomeric species.

e.g. formation of polythene from ethene.

B) Copolymers: - These polymers are formed by the polymerisation of two different monomers, e.g. formation of buna-S from styrene and buta-1,3-diene. Some common addition polymers are polythene, polystyrene, teflon, buna-S, PVC etc.

C) Condensation polymers: - These are formed by repeated condensation rxn b/w two different bi-functional or tri-functional monomeric units, with the elimination of small molecules such as alcohol, water, HCl etc. e.g. nylon-6,6 is formed from hexamethylene diamine and adipic acid.

Some condensation polymers are terylene or dacron, nylon-6, bakelite etc.

On the basis of molecular forces:-

1) Elastomers: - These are rubber-like solids with elastic properties. They have the weakest intermolecular forces. eg:- buna-S, buna-N, Neoprene etc.

2) Fibres: - They have strong intermolecular forces like H-bonding; eg. Polyamides (nylon-6,6), Polyesters (terylene) etc.



3) **Thermoplastic polymers**: - These are linear OR slightly branched long chain molecules having intermediate intermolecular forces of attraction, between elastomers and fibres.

eg: polythene, polystyrene, polyvinyls etc.

4) **Thermosetting polymers**: - These are cross-linked OR heavily branched molecules which when heated in a mould; undergo a permanent change in their chemical composition to give a hard mass.

They cannot be reused, eg: bakelite; urea-formaldehyde resins etc.

**On the Basis of growth polymerisation:** -

- i) Addition polymers.
- ii) Condensation polymers.

## TYPES OF POLYMERISATION REACTIONS

Polymerisation reactions are of two types: -

- 1) Addition polymerisation OR chain growth polymerisation
- 2) Condensation polymerisation OR step growth polymerisation.

### Addition OR chain growth polymerisation: -

When same OR different monomer molecules (unsaturated compounds) successively add together on a large scale to form a polymer, then the type of polymerisation is known as addition OR chain growth polymerisation.

This involves free radical OR ionic mechanism among which free radical one is the most common mode.

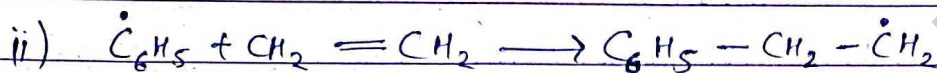
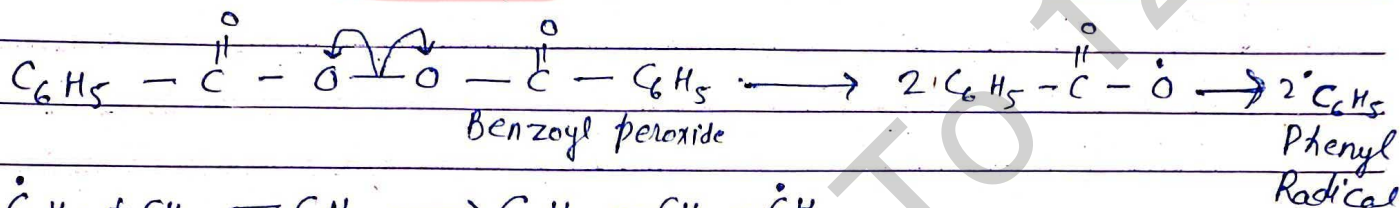


## Free Radical Mechanism: -

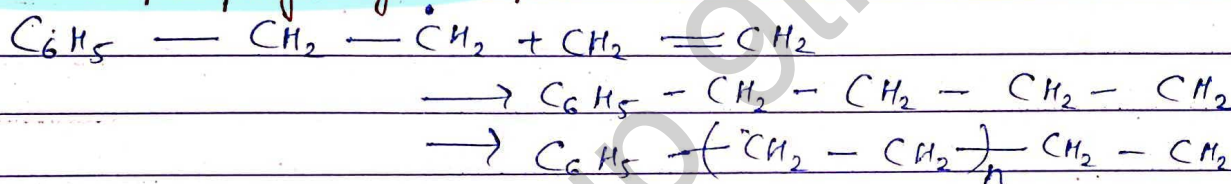
it includes chain initiating step, chain propagating step and chain terminating step. The sequence of steps may be depicted as follows: -

### \* Chain initiation step: -

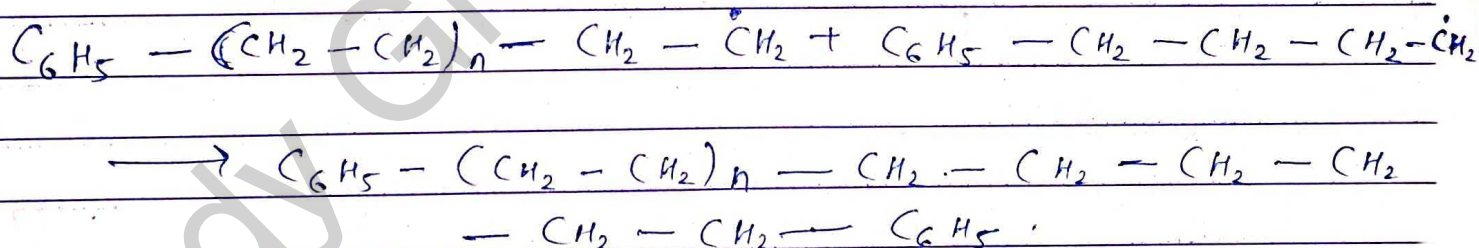
i)



### \* Chain propagating step: -



### \* Chain Terminating step: -



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## Preparation of Polythene by Addition Polymerisation

In the preparation of polythene from ethene, benzoyl peroxide acts as an initiator or free radical generator. Polythene is of two types :

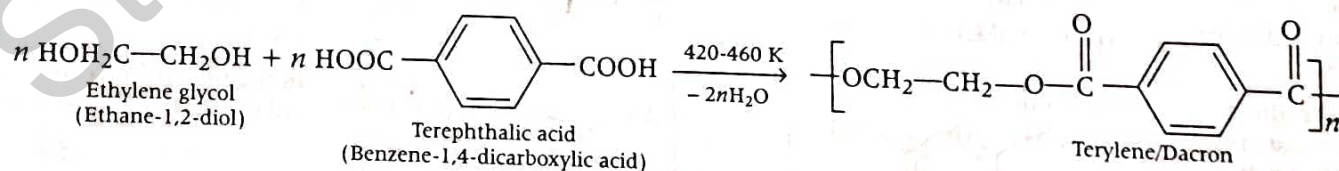
- Low density polythene (LDP)** It is chemically inert and tough but flexible and poor conductor of electricity.
- High density polythene (HDP)** It is chemically inert and more tougher and harder due to close packing.

Some commercially important addition polymers are :

| Name of polymer                  | Monomer                                          | Structure of polymer                                                        | Uses                                                                    |
|----------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Polythene                        | Ethene                                           | $\left[ \text{CH}_2 - \text{CH}_2 \right]_n$                                | Flexible pipes, buckets, toys, bottles, etc.                            |
| Polypropene                      | Propene                                          | $\left[ \text{CH}_2 - \underset{\text{CH}_3}{\text{CH}} \right]_n$          | Manufacture of ropes, toys, pipes, fibres.                              |
| Polystyrene                      | Styrene                                          | $\left[ \text{CH}_2 - \underset{\text{C}_6\text{H}_5}{\text{CH}} \right]_n$ | As insulator, wrapping materials, manufacture of television cabinets. ✓ |
| Polyvinyl chloride (PVC)         | Vinyl chloride                                   | $\left[ \text{CH}_2 - \underset{\text{Cl}}{\text{CH}} \right]_n$            | Manufacture of raincoats, handbags, vinyl flooring, water pipes.        |
| Teflon                           | Tetrafluoroethene<br>$\text{CF}_2 = \text{CF}_2$ | $\left[ \text{CF}_2 - \text{CF}_2 \right]_n$                                | In making oil seals, gaskets and for non-stick surface coated utensils. |
| Polyacrylonitrile (PAN) or orlon | Acrylonitrile<br>$\text{CH}_2 = \text{CHCN}$     | $\left[ \text{CH}_2 - \underset{\text{CN}}{\text{CH}} \right]_n$            | In making clothes, carpets and blankets. ✓                              |

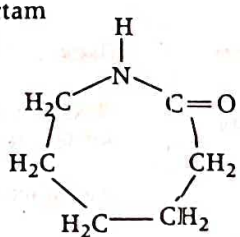
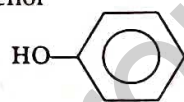
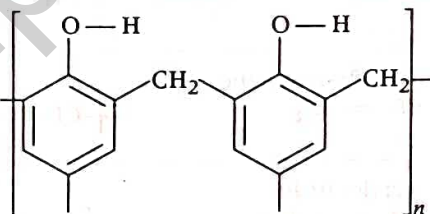
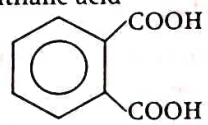
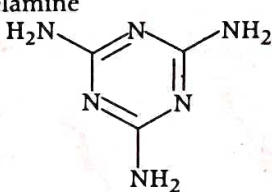
## Condensation or Step Growth Polymerisation

It involves the repetitive condensation reaction between two bi-functional monomers. Some simple molecules like water, alcohol, etc., are lost in these polycondensation reactions and high molecular mass condensation polymers are formed. e.g. terylene, nylon-6,6, nylon-6, phenol-formaldehyde polymers and melamine-formaldehyde polymers, etc.





Some important condensation polymers are:

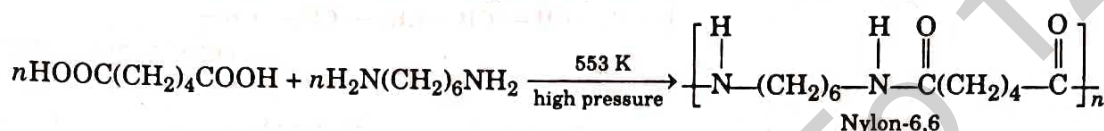
| Polymer                                   | Monomer and their structure                                                                                                                                                          | Structure of polymer                                                                                                      | Uses                                                                           |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Nylon-6,6                                 | (i) Hexamethylene diamine<br>$\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$<br>(ii) Adipic acid<br>$\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$                                       | $\left[ \text{N}-(\text{CH}_2)_6-\text{N}-\text{C}(=\text{O})-(\text{CH}_2)_4-\text{C}(=\text{O}) \right]_n$              | In making sheets, bristles for brushes.                                        |
| Nylon-6                                   | Caprolactam<br>                                                                                     | $\left[ \text{C}(=\text{O})-(\text{CH}_2)_5-\text{N} \right]_n$                                                           | For the manufacture of tyre cords, fabrics and ropes.                          |
| Dacron (terylene) or polyesters           | (i) Ethylene glycol,<br>$\text{HOH}_2\text{C}-\text{CH}_2\text{OH}$<br>(ii) Terephthalic acid<br> | $\left[ \text{O}-\text{CH}_2-\text{CH}_2-\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{C}(=\text{O}) \right]_n$ | In manufacturing tyre cords, seat belts, wash and wear fabrics.                |
| Bakelite                                  | (i) Phenol<br><br>(ii) Formaldehyde, $\text{HCHO}$                                                |                                       | For making combs, electrical switches, handles of utensils and computer discs. |
| Glyptal                                   | (i) Ethylene glycol<br>$\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$<br>(ii) Phthalic acid<br>    | $\left[ \text{OCH}_2-\text{CH}_2\text{OOC}-\text{C}_6\text{H}_4-\text{CO} \right]_n$                                      | Manufacture of paints and lacquers.                                            |
| Urea formaldehyde resin                   | (i) Urea $\text{NH}_2-\text{CO}-\text{NH}_2$<br>(ii) Formaldehyde, $\text{HCHO}$                                                                                                     | $\left[ \text{NH}-\text{CO}-\text{NH}-\text{CH}_2 \right]_n$                                                              | For making unbreakable cups and laminated sheets.                              |
| Melamine formaldehyde polymer or Melamine | (i) Melamine<br><br>(ii) Formaldehyde, $\text{HCHO}$                                              | $\left[ \text{HN}-\text{C}_6\text{H}_3\text{N}_3-\text{NHCH}_2 \right]_n$                                                 | In the manufacture of unbreakable crockery.                                    |

## 1.3 Polyamides

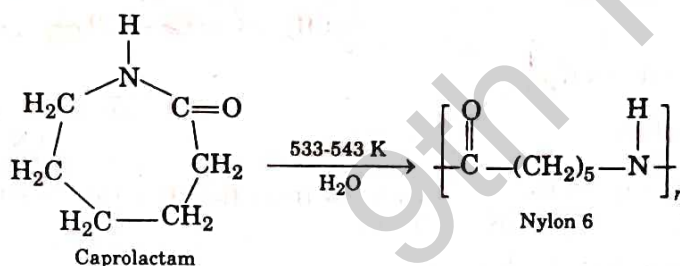
These polymers possessing amide linkages are important examples of synthetic fibres and termed as nylon.

### Preparation of Nylons

- (i) **Nylon-6, 6** It is prepared by the condensation polymerisation of hexamethylenediamine with adipic acid under high pressure and at high temperature.

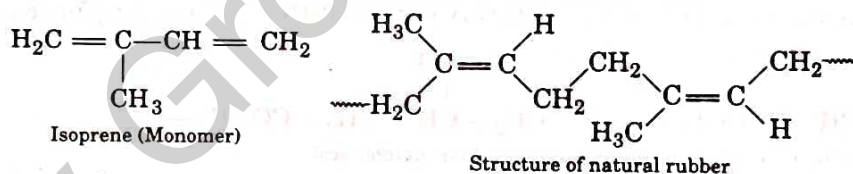


- (ii) **Nylon-6** It is obtained by heating caprolactam with water at a high temperature.



## 1.4 Natural Rubber

Natural rubber is a *cis*-1,4-polyisoprene and possesses elastic properties. It consists of various chains held together by weak van der Waals' interactions and has a coiled structure.



It is used in making footwears, foam mattresses, balloons, toys, rubber bands, etc.

### Vulcanisation of Rubber

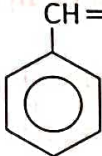
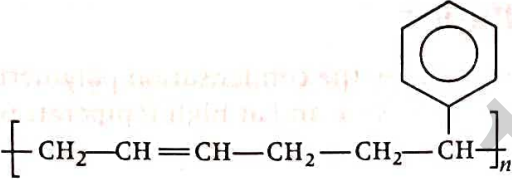
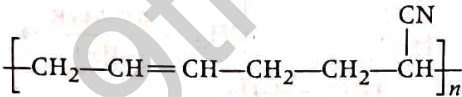
Vulcanisation of rubber is the process of the heating of natural rubber with sulphur and an appropriate additive at a temperature range between 373K to 415K to improve its physical properties. Sulphur forms cross-links at the reactive sites of the double bond and thus rubber gets stiffened.

## 1.5 Synthetic Rubber

Synthetic rubber is any vulcanisable rubber like polymer, which is capable of getting stretched to twice its length. It returns to its original shape and size when the external stretching force is released. This type of rubbers are either homopolymers of 1, 3-butadiene derivatives or copolymers of 1, 3-butadiene or its derivatives with another unsaturated monomers.

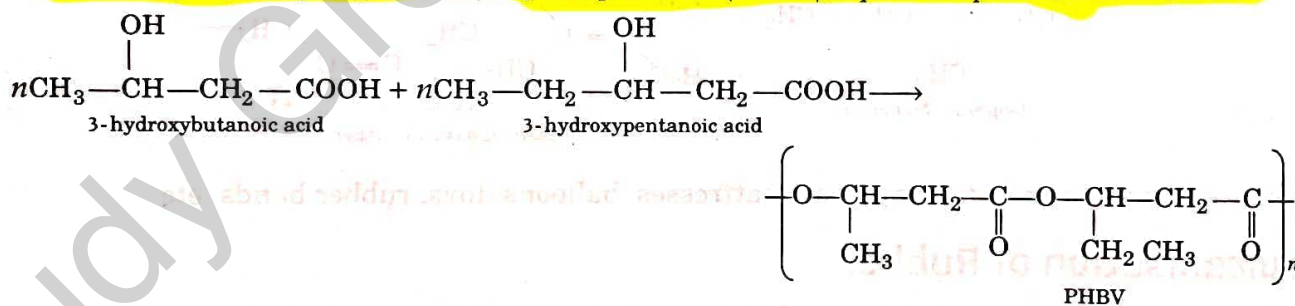


### Some important synthetic rubbers:

| Polymer                                  | Monomer and their structure                                                                                                                                                                                  | Structure of polymer                                                                      | Uses                                                                   |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Buna-S<br>(also an example of copolymer) | (i) 1,3-butadiene,<br>$\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$<br>(ii) Styrene,<br>$\text{CH} = \text{CH}_2$<br> |         | For the manufacture of auto tyres, floor tiles, cable insulation, etc. |
| Neoprene                                 | Chloroprene,<br>$\text{CH}_2 = \underset{\text{Cl}}{\text{C}} - \text{CH} = \text{CH}_2$                                                                                                                     | $\left[ \text{CH}_2 - \underset{\text{Cl}}{\text{C}} = \text{CH} - \text{CH}_2 \right]_n$ | For manufacturing conveyor belts, gaskets and hoses.                   |
| Buna-N                                   | (i) 1,3-Butadiene,<br>$\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$<br>(ii) Acrylonitrile, $\text{CH}_2 = \underset{\text{CN}}{\text{CH}}$                                                             |        | In making seals, tank lining, etc.                                     |

## 1.6 Biodegradable Polymers

The polymers that can be broken down rapidly by enzyme catalysed reaction are called biodegradable polymers. e.g. Poly- $\beta$ -hydroxybutyrate-co- $\beta$ -hydroxyvalerate (PHBV), nylon-2-nylon-6, dextran, etc.



# Thank you so much